

Dear Editors,

We are pleased to submit our manuscript, “Duration judgments with conflicting audiovisual cues,” for consideration as a Research Article in the Journal of Vision.

How the perceptual system resolves conflicting sensory information is a central question in multisensory research. Although Bayesian causal-inference models predict that integration should weaken when cross-modal discrepancies become sufficiently large, whether this occurs in audiovisual duration perception remains unclear. We addressed this question by measuring auditory duration judgments across a large range of audiovisual conflict levels, ranging from -250 to $+250$ ms, and two levels of auditory reliability.

We found that perceived auditory duration shifted systematically toward the task-irrelevant visual duration, with substantially greater visual influence when auditory information was less reliable. Importantly, this influence remained approximately linear across the tested conflict range, providing little evidence for the nonlinear breakdown of integration predicted by causal inference. Computational model comparison favored probabilistic cue switching for the largest number of participants, and model-recovery analyses identified cue-switching data correctly in 84% of simulations. Together, these results provide converging support for probabilistic cue switching while acknowledging some overlap among the candidate models.

We believe the manuscript will be of particular interest to readers of the Journal of Vision because it links visual temporal perception with multisensory integration and uses computational model comparison to distinguish among reliability-weighted fusion, Bayesian causal inference, and heuristic cue-switching accounts. More broadly, our findings show that auditory duration judgments remain strongly influenced by vision across large cue conflicts and support probabilistic cue switching as an alternative to causal inference in temporal perception.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and its submission to the Journal of Vision. The authors declare no competing interests. Data and analysis code are publicly available at: <https://github.com/omeryildiran/av-dur-estimation>.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,
Omer Yildiran
New York University
Dept. of Psychology
omer.yildiran@nyu.edu
On behalf of all authors.